Examination plan and findings: type-I conservatism of the interaction test

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1. Objective

The empirical type-I error of the biomarker-by-treatment interaction test (`bm:Db` under specifications A1 and A3, `bm:Dbc` under A2) runs systematically below the nominal 0.05 across the carryover-sensitivity factorial. The pooled mean over the 540 null cells of the production run is 0.039, and against roughly 270000 null replicates this is on the order of thirty standard errors below nominal. The deflation is therefore real and not a sampling artefact.

The manuscript (§4.6) attributes the conservatism in a single line to 'the conservative AR(1) reference distribution at moderate sample sizes'. That attribution is plausible but unverified. The purpose of this examination is to localise the conservatism to one of four inference channels, attribute it to a mechanism, and produce an evidenced statement that either confirms the §4.6 attribution or replaces it.

A preliminary point has already been settled. At the null (`c_bm = 0`) the two DGP architectures collapse to a bit-identical data-generating process, confirmed by `check-null-architecture-equivalence.R`. The architecture axis therefore carries no information at the null, and the examination need not stratify by architecture.

Sections 2 to 12 record the examination as planned. Sections 13 to 15, added after execution on 2026-05-20, report the findings obtained and the resulting plan to address the conservatism.

2. Stage 0: scope and reference cell

The examination begins at a single null cell and expands only as the evidence requires.

- **Primary null cell.** `c_bm = 0`, exponential DGP, Hybrid design, `N = 70`, `t1half = 1.0`, `rho = 0.7`. This is the null counterpart of the manuscript reference configuration.
- **Replicates.** 5000 for the primary cell, giving a Monte Carlo standard error on the type-I estimate of approximately 0.0031 and resolving 0.039 against 0.050 at roughly 3.5 standard errors. 10000 replicates are held in reserve should the p-value ECDF tail require a cleaner estimate.
- **Reproducibility.** A fixed master seed, with per-replicate seeds derived as in the production driver. The complete per-replicate output is retained, not merely the rejection indicator.

3. Stage 1: instrument the per-replicate output

The production fitter `fit_spec()` in `simulation-core.R` returns only `estimate`, `std_error`, `p_value`, and `converged`. The decomposition that follows requires more. A standalone diagnostic driver, `diagnose-null-type1.R`, will reuse `simulation-core.R` for data generation but supply an augmented fitter that records, per replicate:

- the degrees of freedom `lme` assigns the interaction term, from `summary(fit)$tTable[, 'DF']`;
- the estimated corCAR1 correlation parameter, the random-intercept standard deviation, and the residual standard deviation, from `VarCorr(fit)` and `fit$sigma`;
- the `non_positive_definite_rate` attribute returned by `generate_data()`, so that the positive-definiteness correction can be excluded as a contributor at negligible cost;
- the optimizer convergence messages.

The diagnostic driver does not modify the production pipeline.

4. Stage 2: decompose into the four inference channels

The interaction test is a Wald t-statistic, `T = estimate / std_error`, rejected when `|T|` exceeds the t critical value at the assigned degrees of freedom. Conservatism must enter through one of four channels, and the channels are separable. From the primary 5000-replicate output we compute the following.

1. **Point-estimate bias.** The mean of `estimate` and a test of its equality to zero. The expected value is approximately zero; a non-zero mean would constitute a separate problem and would, if anything, mask the conservatism rather than cause it.
2. **Standard-error calibration.** The ratio of the mean model-reported `std_error` to the empirical standard deviation of `estimate` across replicates, together with the full distribution of `std_error` so that a uniform upward shift can be distinguished from a heavy right tail. A ratio above 1.05 implicates standard-error over-estimation.
3. **Statistic dispersion.** The standard deviation of `T` under the null, which should be approximately one under correct calibration. A value below one confirms a conservative reference.
4. **p-value uniformity.** The empirical CDF of the p-values against the diagonal, Kolmogorov-Smirnov and Anderson-Darling tests against Uniform(0, 1), and the type-I rate at alpha in {0.01, 0.05, 0.10}. Whether all three alpha levels are depressed or only the tail distinguishes a global scale error from a tail-shape error.

The deliverable of this stage is a four-row decomposition table and the p-value ECDF figure.

5. Stage 3: separate the DF channel from the SE channel

The p-values are recomputed from the same `estimate` and `std_error` against a standard-normal reference in place of the t reference.

- If the type-I rate rises to approximately 0.05 under the normal reference, the `lme` degrees-of-freedom rule is the cause.
- If the type-I rate remains at approximately 0.039, the degrees-of-freedom rule is excluded and the cause lies in the standard error or the joint estimate-and-SE calibration. The examination then proceeds to Stage 4.

A prior check short-circuits this stage. The assigned degrees of freedom are tabulated directly; with `N = 70` across eight measurement occasions the level-one degrees of freedom are likely in the hundreds, in which case the t and normal references are numerically indistinguishable and the degrees-of-freedom channel is excluded immediately.

6. Stage 4: attribute the SE mis-calibration to a covariance model

The same saved null datasets are refitted under a ladder of correlation specifications, comparing the interaction standard error, the calibration ratio, and the type-I rate across them.

- **M0.** `lm(Sx ~ bm + t + Db + bm:Db)`, ignoring all within-subject correlation. Expected to be anti-conservative and serving as a lower anchor.
- **M1.** `lme` with a random intercept only, no corCAR1.
- **M2.** `gls` with a corCAR1 residual only, no random intercept.
- **M3.** `lme` with both a random intercept and a corCAR1 residual, the production model.

The data-generating process produces AR(1)-correlated factors plus a participant-level baseline term `BL` that acts as a random intercept. Specification M3 is therefore the well-specified member of the ladder, and M1 and M2 are each mis-specified. The reading is as follows. If M3 alone is conservative while M1 is calibrated, the near-redundant pairing of a random intercept with a corCAR1 residual on only eight occasions is implicated. If M1, M2, and M3 are all conservative, the cause is the generic finite-sample plug-in Wald-t and is not specific to the redundancy.

7. Stage 4b: gold-standard calibration counterfactual (conditional)

This stage is undertaken only if Stages 2 to 4 point to the Wald approximation. The conservatism is then checked against a reference that does not rely on that approximation.

- A parametric-bootstrap or permutation null distribution for the interaction statistic, computed on a subsample of datasets. If those p-values are uniform while the Wald p-values are not, the Wald approximation is conclusively the cause.
- Optionally, an `lmer` fit with a Kenward-Roger small-sample correction by way of `pbkrtest`, accepting that `lmer` cannot carry the corCAR1 residual and so tests only the random-intercept structure.

8. Stage 5: design dependence

The Stage 2 decomposition is repeated for the CO, Hybrid, and OLBDC designs at `c_bm = 0`, `N = 70`. Separately, each design's long-form design matrix is inspected: the correlation and variance inflation factor between `Db` and `t`, and between the `bm:Db` and `bm:t` columns. If the conservatism tracks the `Db-t` collinearity ranking across designs, design-induced collinearity is a contributing channel. If the type-I rate is uniformly near 0.039 irrespective of design, collinearity is excluded.

9. Stage 6: sample-size gradient

The null decomposition is run at `N` in {35, 70, 140, 280}. This is the decisive diagnostic. A finite-sample Wald artefact must attenuate as `N` grows, with the type-I rate approaching 0.05. If it does, the conservatism is benign and expected, and the manuscript may say so with evidence. If the type-I rate remains near 0.039 at large `N`, the cause is asymptotic, constitutes a structural mis-specification, and warrants a correction rather than a footnote.

10. Decision logic

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Stage 2  ratio > 1.05 .................... SE over-estimation
         SD(T) < 1, ECDF bowed ......... conservative reference confirmed
Stage 3  normal reference fixes it ...... DF rule is the cause
         normal reference does not ...... SE/calibration, go to Stage 4
Stage 4  M1 calibrated, M3 not ......... RI + corCAR1 redundancy
         M1, M2, M3 all conservative .... generic finite-sample Wald-t
Stage 5  tracks Db-t collinearity ....... design collinearity contributes
Stage 6  type-I -> 0.05 as N grows ...... finite-sample, benign
         type-I flat in N .............. asymptotic mis-specification
```

11. Deliverables

1. `diagnose-null-type1.R`, the standalone diagnostic driver.
2. A short results note containing the four-channel decomposition table, the p-value ECDF figure, the model-ladder table, the design and sample-size tables, and an attributed one-paragraph conclusion.
3. A concrete recommendation for §4.6 of the manuscript: either confirm the finite-sample attribution with the sample-size gradient as evidence, or replace it.

12. Cost and sequencing

All stages are computationally light. At the throughput observed in the development run, the 5000-replicate primary cell is on the order of a few minutes, and the model ladder and the sample-size gradient multiply that by small constants. The whole examination is minutes of compute, not hours.

The recommended sequencing is to run Stages 1 to 3 first, since they are quick and may already exclude the degrees-of-freedom channel and quantify the standard-error ratio, and then to decide whether Stages 4 to 6 are required. Stages 4b and 5 are conditional; Stage 6 should be run regardless, because it is the single most decisive test of whether the conservatism is the benign finite-sample behaviour the manuscript assumes.

13. Results

The examination was executed on 2026-05-20. Stages 1 to 3 ran on the primary null cell at 5000 replicates; Stage 4 refitted the identical datasets under the model ladder at 5000 replicates; Stage 6 ran the production model M3 across the sample-size grid at 1500 replicates per point. The driver scripts are `diagnose-null-type1.R` for Stages 1 to 3 and `diagnose-null-type1-stages46.R` for Stages 4 and 6.

13.1 Stages 1 to 3: the conservatism is standard-error over-estimation

Spec	mean est	se_ratio	sd(T)	type-I 0.05	type-I 0.05 (z)	mean DF
A1	0.032	1.069	0.932	0.033	0.034	487
A2	0.039	1.097	0.909	0.029	0.029	487
A3	0.033	1.061	0.939	0.035	0.036	486

The interaction estimate is effectively unbiased and the degrees-of-freedom channel is excluded. The standard error is over-estimated by 6 to 10 percent (`se_ratio`), and this fully accounts for the compressed test statistic, since the standard deviation of the statistic equals the reciprocal of `se_ratio` to three digits. Recomputing the p-values against a normal reference moves the type-I rate by approximately 0.001, confirming that the assigned degrees of freedom, approximately 487, are not the cause. The positive-definiteness correction was triggered on none of the four Hybrid paths and is excluded.

A small positive bias in the null interaction estimate is present (mean approximately 0.033, about 0.036 of a standard deviation, roughly 2.5 standard errors from zero). It is model-independent, points toward rejection rather than away from it, and therefore cannot cause the conservatism, but it is recorded here for completeness and noted again in §15.5.

The null p-value ECDF lies uniformly below the diagonal across the whole unit interval, the signature of a global scale compression rather than a tail anomaly.

13.2 Stage 4: the corCAR1 layer inflates the interaction SE

Model	Structure	se_ratio A1	se_ratio A2	type-I A1	type-I A2
M0	lm	1.323	1.289	0.010	0.012
M1	lme, ~1|ptID	0.989	0.972	0.049	0.055
M2	gls, corCAR1	1.023	1.070	0.043	0.036
M3	lme, ~1|ptID + corCAR1	1.068	1.097	0.033	0.029

The ladder is unambiguous. M1, a random intercept with no residual correlation, is well calibrated: its standard-error ratio sits at 0.97 to 0.99 and its type-I rate at 0.049 to 0.055. M3, the production model, adds a corCAR1 residual on top of that random intercept and is conservative: the ratio rises to 1.07 to 1.10 and the type-I rate falls to 0.029 to 0.033. The corCAR1 residual-correlation layer is therefore the proximate cause of the conservatism.

Two ladder results merit comment. First, M0, ordinary least squares, is not anti-conservative as Stage 4's plan text anticipated but strongly conservative, with a standard-error ratio near 1.3. The reason is that the interaction is a within-subject contrast, and ordinary least squares charges it the full residual variance, between-subject plus within-subject, when the relevant error is only the within-subject part. The plan's expectation for M0 was therefore wrong, and the corrected reading is recorded here. Second, M2, a corCAR1 residual with no random intercept, is intermediate. Taken together, the ladder shows that the random intercept alone recovers calibration and the corCAR1 addition removes it.

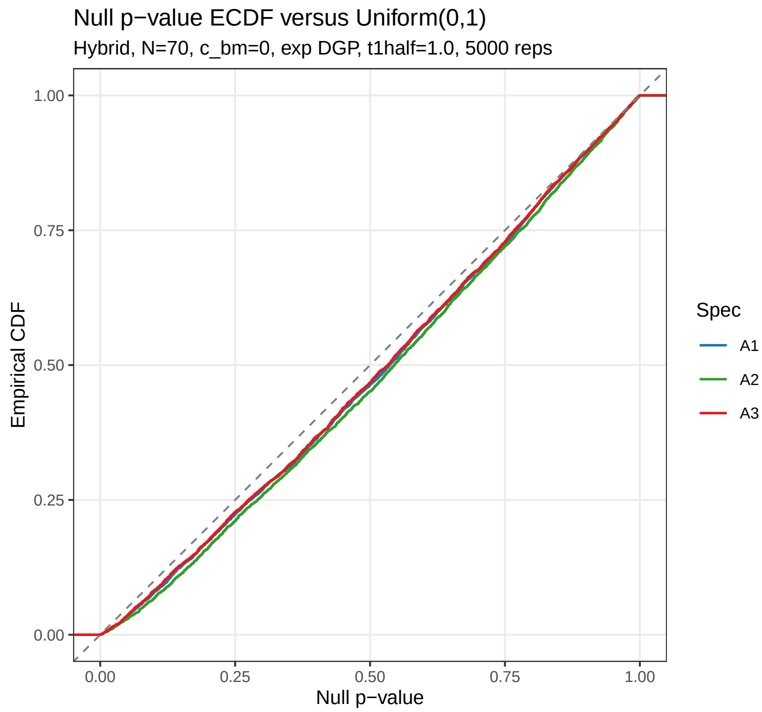

Figure 1: Stage 2: null p-value empirical CDF against Uniform(0,1) for the primary null cell. All three specifications bow uniformly below the diagonal; A2 is the lowest curve, the most conservative.

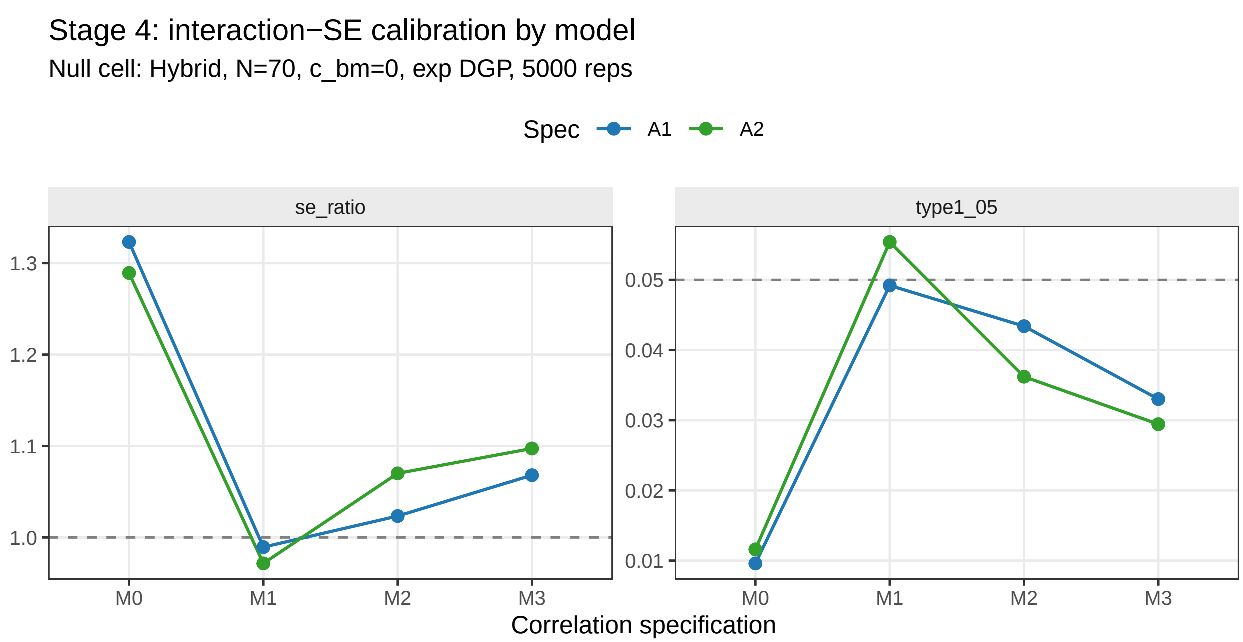

Figure 2: Stage 4: interaction standard-error calibration across the model ladder. The left panel is the ratio of model-reported SE to the empirical SE, the right panel the realised type-I rate. M1 alone sits on the reference lines; M3 does not.

13.3 Stage 6: the conservatism is asymptotic, not finite-sample

N	se_ratio A1	se_ratio A2	se_ratio A3	type-I A1	type-I A2	type-I A3
35	1.072	1.103	1.064	0.038	0.028	0.039
70	1.056	1.083	1.050	0.043	0.031	0.045
140	1.086	1.098	1.079	0.041	0.033	0.042
280	1.052	1.070	1.046	0.037	0.028	0.035

This is the decisive result. The plan posited that a finite-sample Wald artefact would attenuate as N grows, with the type-I rate approaching 0.05. It does not. Across N from 35 to 280 the standard-error ratio holds at 1.05 to 1.10 with no trend toward one, and the type-I rate holds at 0.028 to 0.045 with no trend toward 0.05. At N = 280 the production specification A2 still rejects at 0.028. The conservatism is a structural feature of the M3 model and does not vanish with sample size.

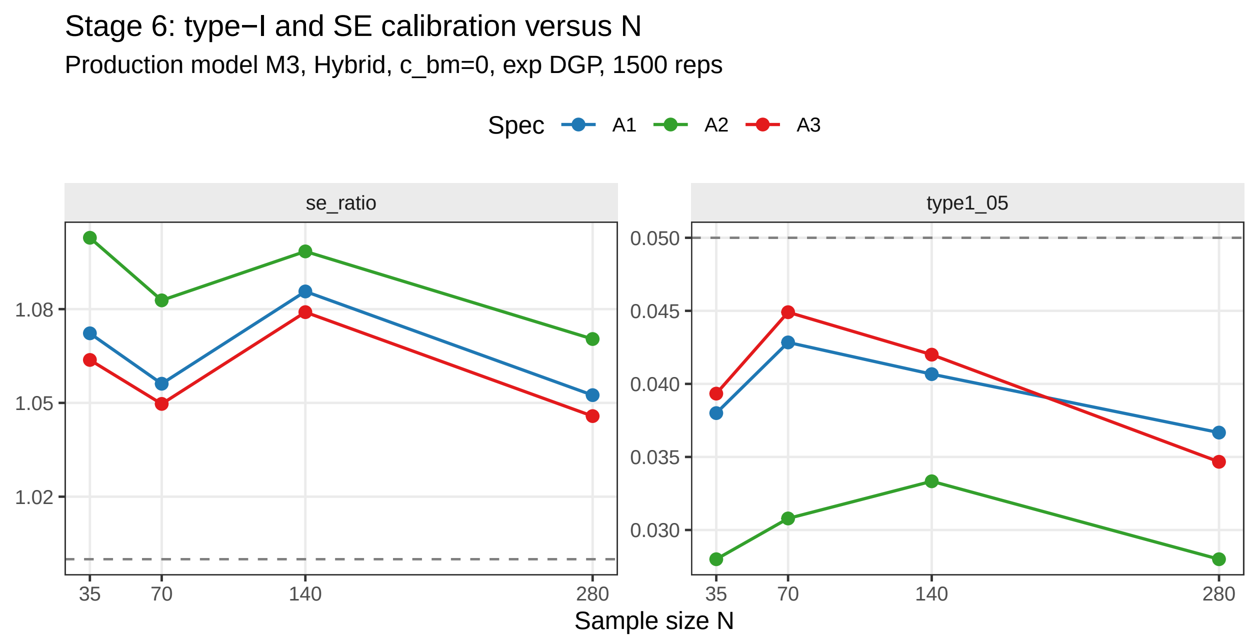

Figure 3: Stage 6: type-I rate and standard-error ratio against sample size under the production model M3. Neither panel trends toward its reference line as N grows.

The explanation consistent with all three stages is as follows. The simulated symptom score is a mixture of three AR(1)-correlated factors with timepoint-dependent variances, plus a participant-level baseline term. The M3 working covariance, a single random intercept plus a single-parameter corCAR1 residual, cannot match that structure exactly. With only eight measurement occasions per participant the random-intercept variance and the corCAR1 correlation parameter are weakly separated, and the resulting model-based standard error for the interaction is biased upward. Increasing the number of participants sharpens the variance-parameter estimates but does not change the eight-occasion structure that drives the mismatch, so the bias persists. This is why the conservatism is asymptotic.

14. Conclusions

1. The type-I conservatism is real and is standard-error over-estimation. The model-reported standard error of the interaction coefficient exceeds its true sampling standard deviation by 6 to 10 percent, the test statistic is compressed by the reciprocal of that factor, and the null p-values are stochastically too large at every alpha level.

2. The cause is the corCAR1 residual-correlation term in the production analysis model. A random intercept alone (M1) is well calibrated; adding corCAR1 (M3) inflates the interaction standard error. The effect is largest for the exposure-weighted specification A2, which is consequently the most conservative, consistent with A2 sitting lowest in the manuscript's Figure 4.
3. The conservatism is asymptotic, not a moderate-sample-size artefact. It does not attenuate from N = 35 to N = 280. The manuscript's §4.6 attribution, 'the conservative AR(1) reference distribution at moderate sample sizes', is correct in naming the AR(1) term but wrong in implying the effect is finite-sample and would diminish with N.
4. The degrees-of-freedom rule, the positive-definiteness correction, and point-estimate bias are all excluded as causes.
5. The conservatism is in the safe direction. It is a valid test that under-rejects, not an inflated test. Its practical cost is a modest loss of power, which means the manuscript's power estimates are mildly pessimistic rather than optimistic. The headline ranking A2 greater than A1 approximately equal to A3 is not threatened, because all three specifications are inflated by similar factors and a recalibration would lift all three powers while preserving their order.

15. Plan to address the conservatism

The conservatism is structural, so it will not be removed by more replicates or larger N. Four remedies are available, listed from the most principled to the most minimal.

15.1 Option A: cluster-robust standard errors (recommended)

Retain the M3 point estimate and the corCAR1 model, which the project adopts for clinical realism, but replace the model-based standard error of the interaction with a cluster-robust sandwich standard error, clustering on `ptID`. A sandwich estimator is consistent for the true sampling variance whether or not the working covariance is correctly specified, so it removes the asymptotic bias identified in Stage 6. The bias-reduced CR2 variant in the `clubSandwich` package is appropriate, since it carries a small-sample correction and supports `lme` objects. This is the recommended fix: it preserves the analysis model and corrects only the inference.

15.2 Option B: drop the corCAR1 term

Use M1, a random intercept only. Stage 4 shows M1 is well calibrated. The cost is the loss of the corCAR1 residual structure, which the project adopted deliberately, and which M1 mis-specifies by treating the within-subject residual as independent. M1's fixed-effect test is nonetheless calibrated, so this option is defensible, but it is a larger change to the analysis than Option A and should not be taken without re-running the principal factorial.

15.3 Option C: parametric-bootstrap reference

Replace the Wald reference for the interaction test with a parametric-bootstrap or permutation null distribution. This is the gold standard for calibration and is robust to the working-covariance mismatch, but it is computationally heavy across a 540-cell factorial and is best reserved for confirmation on selected cells rather than as the production default.

15.4 Option D: document without changing the analysis

If the analysis is not to be re-run, the minimum acceptable response is to correct §4.6. The conservatism should be described as a structural consequence of the corCAR1 term, persistent at all sample sizes, of magnitude 6 to 10 percent in standard-error terms and roughly 0.03 to 0.04 in realised type-I error, and in the safe direction. The phrase 'at moderate sample sizes' should be removed, since Stage 6 shows the effect does not depend on N.

15.5 Recommended sequence

1. Implement Option A on the primary null cell and confirm that the CR2 cluster-robust standard error brings the standard-error ratio to approximately one and the type-I rate to approximately 0.05.
2. If confirmed, apply the CR2 standard error across the Tier 1 factorial and regenerate the affected summaries. The point estimates and the power ranking are not expected to change; the type-I columns and the absolute power values will.
3. Revise §4.6 of the manuscript per Option D in all cases, since the current 'moderate sample sizes' wording is incorrect regardless of which remedy is adopted.
4. Record the small positive null bias of the interaction estimate (§13.1) in the manuscript's limitations, noting it is negligible in magnitude and does not affect the conclusions.

The first step is a short, single-cell verification, which can be written as `diagnose-null-type1-cr2.R` and run on request.

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